PO12Q - Quantitative Political Analysis:
Uncovering Relationships

Week 10 - Worksheet Solutions

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Exercises with the app

1. Sampling variability and the true parameter
   a) The three values of $\hat{\beta}_1$ will differ from each other and from the true value of 1. Each simulation draws a fresh random sample, so the estimate varies due to sampling variability – the error term ε is different each time.
   b) With n = 1000 the estimates cluster much more tightly around β_1 = 1. Larger samples give OLS more information, reducing the standard error of the estimator.

> **Takeaway:** OLS estimates are random variables. They scatter around the true β_1, but larger samples bring them closer. This is the difference between an estimator (the procedure) and an estimate (one realisation).

2. What heteroscedasticity does – and doesn't – break
   a) $\hat{\beta}_1$ does not shift systematically; it stays centred on the true value. Heteroscedasticity does not bias OLS coefficients – they remain unbiased.
   b) When the Breusch-Pagan test rejects, the OLS standard errors are unreliable because they assume constant variance. The HC3 (robust) standard errors are valid because they do not rely on the homoskedasticity assumption. In practice, the HC3 SEs are typically larger, making inference more conservative but more honest.

> **Takeaway:**
> Heteroscedasticity breaks the formula for standard errors, not the coefficients themselves. Conventional OLS SEs assume constant variance, so when that fails, confidence intervals and p-values become unreliable. Robust (HC3) SEs fix this.

3. The cost of ignoring nonlinearity
   a) The linear fit cuts through the curve, systematically over-predicting y in the middle range of x and under-predicting at the extremes (or vice versa, depending on the sign of β_2). The residuals show a clear U-shaped or inverted-U pattern.
   b) The F-test p-value drops below conventional thresholds (0.05, 0.01) once β_2 is large enough for the curvature to be detectable given the sample size. Fitting a straight line to curved data produces biased predictions – especially at the tails of x, where extrapolation errors compound.

Takeaway:
Unlike heteroscedasticity, nonlinearity biases OLS estimates. The slope $\hat{\beta}_1$ no longer estimates a meaningful quantity because the linear model is the wrong functional form. Residuals carry systematic patterns, and predictions at the extremes of x can be badly wrong.

4. Why collinearity makes you uncertain, not wrong
   a) As ρ rises toward 1, VIF increases sharply (e.g. exceeding 10 or even 100), and standard errors for both x_1 and x_2 inflate correspondingly. The estimates become very imprecise.
   b) The estimate of β_1 changes when x_2 is dropped because x_1 absorbs the effect of the omitted correlated variable. If x_2 is a genuine confounder, only the model including both regressors gives a coefficient with a ceteris paribus interpretation – but the price is large standard errors.
   c) Yes. At high ρ, $\hat{\beta}_1$ and $\hat{\beta}_2$ swing wildly from one simulation to the next, even though on average they remain centred on their true values.

Takeaway:
Collinearity inflates variance but does not introduce bias – OLS is still unbiased, just imprecise. The real danger is instability: estimates swing wildly across samples because the data cannot distinguish each variable's separate contribution. This is a problem of information, not of the estimator.

5. Connecting violations to the Gauss-Markov conditions

Table 1: Mapping of Violations to the BLUE Properties

	Heteroscedasticity	Nonlinearity	Collinearity
Which assumption is violated?	(3) Homoscedasticity: the error variance changes with $\boldsymbol{x}$, so it is no longer constant.	(1) Functional form: the true relationship is non-linear, but we fitted a linear model.	(6) Values of $\boldsymbol{X}$: $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are near-perfectly correlated, so $\boldsymbol{X}'\boldsymbol{X}$ is close to singular.
Is OLS still linear?	Yes: OLS remains a linear function of $\boldsymbol{y}$.	Yes: OLS is still linear in $\boldsymbol{y}$, even though the model is misspecified.	Yes: OLS is still linear in $\boldsymbol{y}$.
Is OLS still unbiased?	Yes: $E(\boldsymbol{\epsilon}\mid\boldsymbol{X}) = 0$ still holds, so OLS remains unbiased.	No: the omitted curvature is absorbed into $\boldsymbol{\epsilon}$, so $E(\boldsymbol{\epsilon}\mid\boldsymbol{X}) \neq 0$, biasing $\hat{\beta}_1$.	Yes: all assumptions for unbiasedness hold; the problem is precision, not bias.
Is OLS still efficient?	No: OLS is no longer minimum variance; GLS or WLS would be more efficient.	No: the model is misspecified, so efficiency is not meaningful.	Technically yes, but the minimum achievable variance is very large due to near-collinearity inflating $\mathrm{Var}(\hat{\boldsymbol{\beta}})$ via VIF.
Are SEs correctly estimated?	No: the standard SE formula assumes $\sigma^2\boldsymbol{I}$, which fails here; use robust (HC) SEs.	No: the model is wrong, so both coefficients and SEs are unreliable; fix the functional form first.	Yes in principle, but they are very large because near-collinearity inflates $\mathrm{Var}(\hat{\boldsymbol{\beta}})$ via VIF.

> **Takeaway:** Different violations break different BLUE properties. Heteroscedasticity leaves OLS unbiased but inefficient with wrong SEs. Nonlinearity breaks unbiasedness entirely. Collinearity preserves unbiasedness and valid SEs, but inflates variance — OLS is still BLUE, just imprecise.

Testing the Classical Linear Assumptions

see RScript.